6-6 The speed of clocks in a gravitational field

Now we want to use the principle of equivalence for figuring out a strange thing that happens in a gravitational field. We'll show you something that happens in a rocket ship which you probably wouldn't have expected to happen in a gravitational field. Suppose we put a clock at the "head" of the rocket ship—that is, at the "front" end—and we put another identical clock at the "tail," as in Figure 6-16. Let's call the two clocks *A* and *B*. If we compare these two clocks when the ship is accelerating, the clock at the head seems

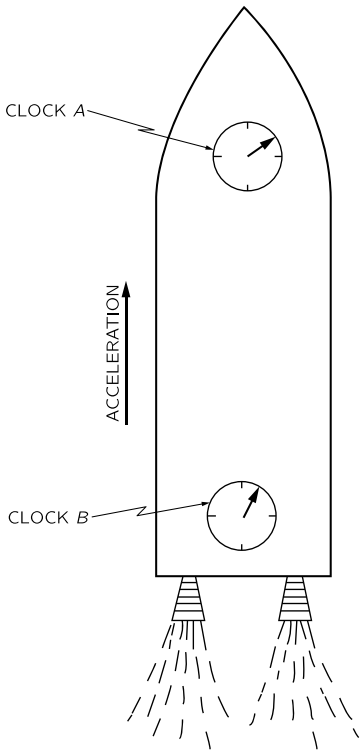

Figure 6-16 An accelerating rocket ship with two clocks.